Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3411-WE01

Title: (BEng/MEng)

Solid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Materials Permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Write your answers in the white-covered answer booklet with barcodes. Begin your answer to each question on a new page. All relevant workings must be shown.

Revision:	

Q1

The consultant company that you work for has won a bid to design a suspended bridge over the Tyne River in Newcastle upon Tyne, in which the bridge deck is hung from two towers on the two sides of the river by steel cables. The cables are connected to the deck using steel plates and bolts, like the one shown below. The total length of the bridge is 2km, and 20 cables from each tower support the bridge deck. The design dead load is $4887\,\text{kg/m}^2$ (which is equal to the **mass** of the bridge components), and the design live load is $2500\,\text{kg/m}^2$ (which is equal to the **mass** of the vehicles that pass the bridge), and the width of the deck is 20m. Assume the dead and live loads are evenly distributed over the bridge deck.

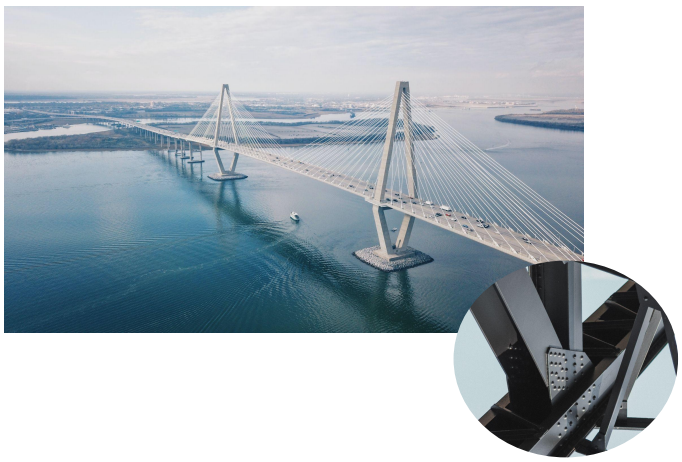


Figure Q1: Suspended bridge system and connections (by David Martin and Josh Datsu)

(a) Determine the total load carried by each cable, considering the dead and live load of the bridge deck, which are uniformly distributed along the width and length of the bridge. **[10%]**

(b) The cables are connected to the deck using carbon steel plates with a thickness of 150mm, which, due to the manufacturing imperfection, may have surface flaws of $a = 5$mm deep. Determine the minimum acceptable width of these steel plates, considering the tensile yield strength of $650\,\text{MPa}$ and the fracture toughness of $52\,\text{MPa}\sqrt{\text{m}}$ for the steel plates. **[20%]**

(c) The plates are connected using 100 structural bolts with a diameter of 38mm, which are characterised as ASTM A490. The tensile strength of these bolts is $1040\,\text{MPa}$, and their fracture toughness is $80.9\,\text{MPa}\sqrt{\text{m}}$. Fastening the bolts for joining the two steel plates applies the tensile stress of $230\,\text{MPa}$ to the bolts, and the dead and live loads of the bridge are transferred between the two plates as shear stress in these bolts. Due to manufacturing imperfections, these bolts may have cracks in any given direction; hence, they go through advanced ultrasonic tests to spot and measure any cracks in these bolts before being used in construction. Given the fact that these cracks may exist in any given direction and the expected tensile and shear stresses, determine the critical crack length of these bolts for inspection purposes. **[35%]**

(d) The steel plates are subject to cyclic loads. During rush hour, when the bridge is fully used, the live load of $2500\,\text{kg/m}^2$ is cyclically loaded and unloaded to each cable connection when the cars pass the cable connection location. The dead load is held constant. The tensile yield strength of the steel plates is $650\,\text{MPa}$ and their fatigue limit is $300\,\text{MPa}$. Use the Soderberg Criterion and calculate the minimum width of the steel plates to avoid fatigue failure. The holes drilled in these plates apply the modified stress concentration factor of $K_f = 2$ and the factor of safety of 2 considered for design purposes. **[35%]**

continued

Q2

A common type of loading in finite element analysis is an imposed surface traction, also know as an inhomogenous Neumann boundary condition.

(a) Why is it important to avoid point loads in finite element analysis and instead represent the load as a distributed surface traction? **[10%]**

Consider the plane strain finite element mesh shown in Figure Q2. The left of the figure shows the global finite element mesh which comprises a single quadratic triangular element. The local system for this element is shown on the right of the figure. Note that the local position of local Node 4 is $\xi = \eta = 0.5$.

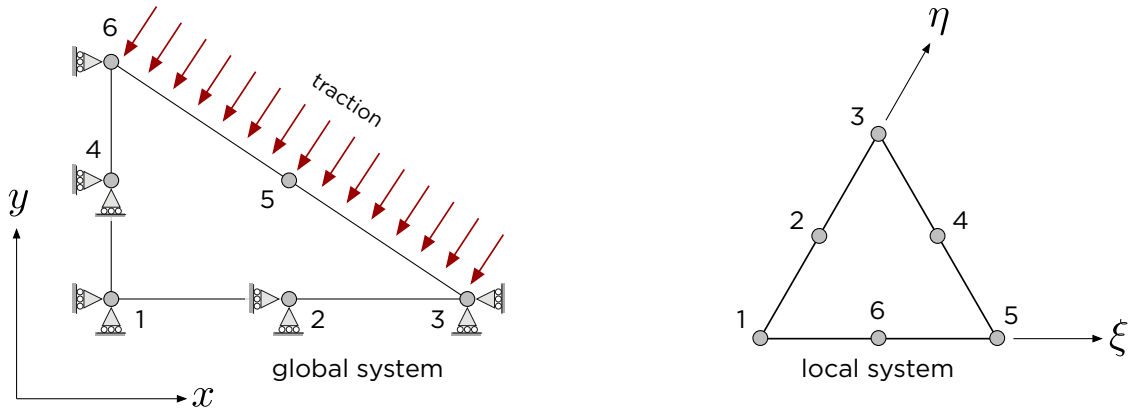


Figure Q2: Global finite element mesh and local element system where $\xi, \eta \in [0, 1]$.

The shape functions for this element are

$$N_1 = (1 - 2\xi - 2\eta)(1 - \xi - \eta), \qquad N_2 = 4\eta(1 - \xi - \eta), \qquad N_3 = \eta(2\eta - 1),$$

$$N_4 = 4\xi\eta, \qquad N_5 = \xi(2\xi - 1) \qquad \text{and} \qquad N_6 = 4\xi(1 - \xi - \eta).$$

The global coordinates of the nodes are

$$\{x\}^T = \{0.0 \quad 3.0 \quad 6.0 \quad 0.0 \quad 3.0 \quad 0.0\} \text{ m} \qquad \text{and} \qquad \{y\}^T = \{0.0 \quad 0.0 \quad 0.0 \quad 2.0 \quad 2.0 \quad 4.0\} \text{ m}.$$

The element is subjected to a constant traction with a magnitude of 10kN/m normal to the edge defined by global nodes 3-5-6, as shown in Figure Q2.

(b) Show that the non-zero equivalent nodal forces associated with the imposed traction are

$$\{f_3\} = \frac{10}{3}\begin{Bmatrix} -2 \\ -3 \end{Bmatrix} \text{ kN}, \qquad \{f_5\} = \frac{40}{3}\begin{Bmatrix} -2 \\ -3 \end{Bmatrix} \text{ kN} \qquad \text{and} \qquad \{f_6\} = \frac{10}{3}\begin{Bmatrix} -2 \\ -3 \end{Bmatrix} \text{ kN}.$$

Show all of your steps and explain any assumptions. **[50%]**

continued

The local stiffness matrix (in MN/m) of the element is

$$[k^e] = \begin{bmatrix} 0.68 & 0.35 & -0.42 & -0.19 & 0.10 & 0.05 & 0.00 & 0.00 & 0.12 & 0.07 & -0.49 & -0.28 \\ 0.35 & 0.97 & -0.28 & -1.11 & 0.07 & 0.28 & 0.00 & 0.00 & 0.05 & 0.05 & -0.19 & -0.19 \\ -0.42 & -0.28 & 1.82 & 0.46 & -0.42 & -0.19 & -0.99 & -0.46 & 0.00 & 0.00 & 0.00 & 0.46 \\ -0.19 & -1.11 & 0.46 & 2.59 & -0.28 & -1.11 & -0.46 & -0.37 & 0.00 & 0.00 & 0.46 & 0.00 \\ 0.10 & 0.07 & -0.42 & -0.28 & 0.31 & 0.00 & 0.00 & 0.28 & 0.00 & -0.07 & 0.00 & 0.00 \\ 0.05 & 0.28 & -0.19 & -1.11 & 0.00 & 0.83 & 0.19 & 0.00 & -0.05 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & -0.99 & -0.46 & 0.00 & 0.19 & 1.82 & 0.46 & 0.00 & 0.28 & -0.83 & -0.46 \\ 0.00 & 0.00 & -0.46 & -0.37 & 0.28 & 0.00 & 0.46 & 2.59 & 0.19 & 0.00 & -0.46 & -2.22 \\ 0.12 & 0.05 & 0.00 & 0.00 & 0.00 & -0.05 & 0.00 & 0.19 & 0.37 & 0.00 & -0.49 & -0.19 \\ 0.07 & 0.05 & 0.00 & 0.00 & -0.07 & 0.00 & 0.28 & 0.00 & 0.00 & 0.14 & -0.28 & -0.19 \\ -0.49 & -0.19 & 0.00 & 0.46 & 0.00 & 0.00 & -0.83 & -0.46 & -0.49 & -0.28 & 1.82 & 0.46 \\ -0.28 & -0.19 & 0.46 & 0.00 & 0.00 & 0.00 & -0.46 & -2.22 & -0.19 & -0.19 & 0.46 & 2.59 \end{bmatrix}$$

where the entries in the matrix are ordered according to the local system shown in Figure Q2. The link between the global element and the local ordering is defined by the following topology vector

$$\text{element topology} = [1 \quad 4 \quad 6 \quad 5 \quad 3 \quad 2].$$

(c) Given the constraints imposed on the element by the boundary conditions shown in Figure Q2, determine the unknown nodal displacements due to the imposed traction. **[40%]**

continued

Q3

Quadratic finite elements are widely using in engineering practice as they represent a good compromise between predictive capability and computational cost. They can also have interesting, and potentially highly effective, properties depending on the global position of the nodes on the element. This question focuses on a *quarter point element* that takes advantage of these properties for fracture/fatigue analysis.

Consider the finite element shown in Figure Q3, which has the following shape functions

$$N_1 = \frac{1}{2}(\xi^2 - \xi), \qquad N_2 = 1 - \xi^2 \qquad \text{and} \qquad N_3 = \frac{1}{2}(\xi^2 + \xi).$$

The global position of Nodes 1 and 3 are

$$x_1 = 0 \text{ m} \qquad \text{and} \qquad x_3 = 1.0 \text{ m}.$$

The global position of Node 2 varies through the question and is specified below as required.

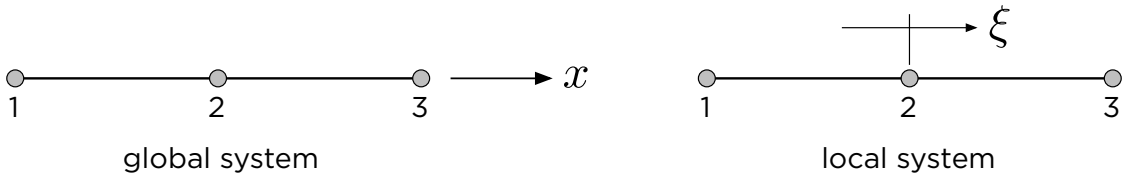


Figure Q3: one dimensional quadratic element, where $\xi \in [-1.1]$

(a) Determine the derivatives of the above shape functions with respect to the local coordinate. **[10%]**

(b) Plot the variation of $\frac{\partial \xi}{\partial x}$ with $x \in [x_1, x_3]$ for: (i) $x_2 = 0.5$m and (ii) $x_2 = 0.25$m. Include appropriate axis limits and label key values. **[50%]**

(c) Comment on the shape of the plot for $x_2 = 0.25$m. What consequence does this have for the stress variation through the element? Why might this be useful for fatigue analysis? **[15%]**

The element with $x_2 = 0.25$m is subjected to a constant body force of $f^b = 1.25\text{kN/m}^3$.

(d) Determine the equivalent nodal force for Node 2 assuming that the element has a constant cross sectional area of A. **[25%]**

ENGI3411-WE01 Solid Mechanics 3
Course: Fracture and Fatigue
Formula sheet

Stress and displacement distribution locally around the tip of a mode I crack oriented in the x-direction

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2} + ...$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa - 1) + \sin^2\frac{\theta}{2}\right] + \cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa + 1) - \cos^2\frac{\theta}{2}\right] + \cdots$$

$\kappa = 3 - 4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

Mode I stress intensity factor

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length a in relatively large plate: $Y = 1.12$

Plastic zone correction

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

Failure diagrams for mean stress correction

Modified Goodman:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1 - \left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_{yield}}\right]$$

Strain-life method

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean} + \frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'^2}{E}(2N_f)^{2b} + \sigma_f'\varepsilon_f'(2N_f)^{b+c}$$

continued

Macro crack growth laws based on fracture mechanics approach

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m\sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

continued

ENGI3411-WE01 Solid Mechanics 3
Course: Finite-element method
Formula sheet

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

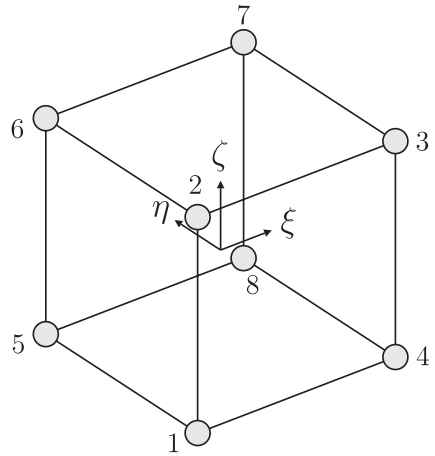
continued

Plane-stress/strain strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{array}\right\} = \left[\begin{array}{cccc}\frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0\\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y}\\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x}\end{array}\right] \left\{\begin{array}{c}u_1\\ v_1\\ \vdots\\ u_{n_{en}}\\ v_{n_{en}}\end{array}\right\}.$$

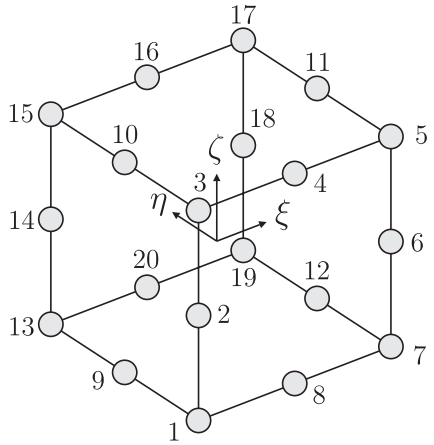
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8\\
N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8\\
N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8\\
N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8\\
N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8\\
N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8\\
N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8\\
N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8
\end{aligned}$$

Tri-quadratic twenty-noded hexahedral



$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8\\
N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4\\
N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8\\
N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4\\
N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8\\
N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4\\
N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8\\
N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4\\
N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4\\
N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4\\
N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4\\
N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4\\
N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8\\
N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4\\
N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8\\
N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4\\
N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8\\
N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4\\
N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8\\
N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4
\end{aligned}$$

continued

Axi-symmetric strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{nen}}{\partial r} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{nen}}{\partial z} \\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{nen}}{r_{nen}} & 0 \\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{nen}}{\partial z} & \frac{\partial N_{nen}}{\partial r} \end{bmatrix} \begin{Bmatrix} u_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}.$$

Global shape function derivatives

$$\begin{Bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial z} \end{Bmatrix} = \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \\ \frac{\partial N_i}{\partial \zeta} \end{Bmatrix}.$$

The Jacobian matrix

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{n_{en}}}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \eta} \\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \zeta} \end{bmatrix} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}} \end{bmatrix}$$

Weak form of equilibrium

$$\int_V [B]^T [D^{\mathsf{e}}][B] dV \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Element stiffness and force vector

$$[k^e] = \int_V [B]^T [D^{\mathsf{e}}][B] dV \qquad \text{and} \qquad \{f^e\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x) dx \approx \sum_{i=1}^{n_{Gp}} f(x_i) w_i$$

n_{Gp}	Abscissae, x_i	Weights, w_i
1	0	2
2	$\pm 1/\sqrt{3}$	1
3	0	$8/9$
	$\pm\sqrt{3/5}$	$5/9$
4	$\pm\sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}}}$	$(18+\sqrt{30})/36$
	$\pm\sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}}}$	$(18-\sqrt{30})/36$

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T [D^{\mathsf{e}}][B]_i w_i \det([J]_i)$$

END